# Mad Scientist 1

Four conditional model results above event COP 1.15

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2026-10-09

**Abstract**

Four declared event models give simulated receiver work above 1.15 times their modelled starting energy and gross external event input. Representative COPs are 1.18827 for a falling-weight lever, 1.57080 for a third-derivative contact law, 17.92044 for a signed-residue port, and 1.67992 for an unconstrained fitted port. We executed 201 new integrations with independent methods, work quadrature, endpoint accounts and separate confirmation cases. Prospective instrument budgets retain an above-threshold margin. These results distinguish additional gravitational descent, hypothetical internal laws and an approximation that disagrees with its passive reference.

PDF created: 2026-10-11 09:26:39 UTC
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## Four predictions to investigate

The target is **event COP greater than 1.15**. Divide net useful receiver work by modelled prepared energy above a declared reference, plus every specified gross external input during the event. Report endpoint stores separately. Preparation efficiency, reset and unknown internal component accounts remain explicit further questions.

The following order gives priority to direct measurement and a clear model interpretation, then to the larger electrical predictions. It is an investigation ranking, not an ordering of hardware success probabilities.

1. **Falling-weight lever:** a loaded descent delivers more work than the initial height reference, with the extra descent recorded. [OP-CT-HW-01]
2. **Third-derivative contact:** an assumed internal force sustains resistance work through the first stop. [OP-LR34-02]
3. **Signed-residue port:** a specified active terminal law drives a receiver and grows the remaining electrical stores. [OP-LR15-01]
4. **Unconstrained fitted port:** a finite-band approximation predicts work that its supplied passive reference does not. [OP-LR35-02]

Each treatment develops a stated model into an executed event study. The last three use hypothetical internal laws whose physical energy accounts remain open. The fitted law's result applies to that law; its reference is calculated alongside it. These are concrete signals and model differences to investigate, with no measured hardware performance implied.

## 1 The falling-weight lever

A teeterboard turns downward arrivals into an upward launch. Timing, leg compression and board flexibility change the outgoing motion. The question begins with an endpoint comparison, then asks what work the complete contact trajectory actually delivers to a receiver.

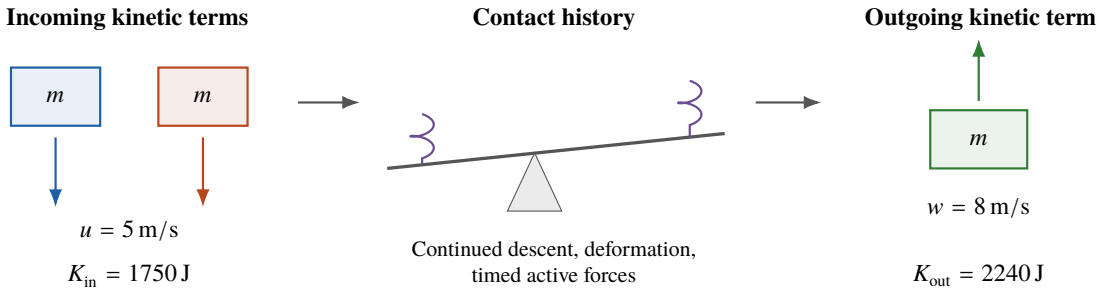



**Figure 1.** Illustrative equal-mass launch comparison. Continued descent and deformation connect the incoming and outgoing kinetic terms.

For two incoming masses and one outgoing mass, each 70 kg, speeds of 5 and 8 m/s give $K_{\text{in}} = 1750$ J and $K_{\text{out}} = 2240$ J: ratio 32/25. Gravity, compliance and active forces can contribute during contact. An ideal fixed pivot supplies impulse at zero work. The trajectory is therefore the interesting missing measurement.

### 1.1 A calculated loaded descent

Our adaptation has two 1 kg vertical drivers, contact arms $a = -0.5$ m, lever inertia $I = 0.15\,\text{kg}\,\text{m}^2$, torsion stiffness 50 N m/rad and bearing damping 0.1 N m s. The receiver arm is $a_R = 0.8$ m. With upward heights $y_i$, angle $\phi$, compression $\delta_i = [a\sin\phi - y_i]_+$ and contact force $f_i = k\delta_i$, Newton's equations during the free event are

$$\begin{aligned} m\ddot{y}_i &= -mg + f_i, \\ I\ddot{\phi} &= -\sum_i f_i a\cos\phi - 50\phi - 0.1\dot{\phi} - b(a_R\cos\phi)^2\dot{\phi}. \end{aligned} \tag{1}$$

Here $g = 9.81\,\text{m/s}^2$, $v_R = a_R\cos\phi\,\dot{\phi}$ and the external viscous receiver takes $P_R = bv_R^2$.
The event starts with simultaneous release from $y_1 = y_2 = 0.30$ m, $\phi = 0$ and zero velocities. It ends 1.5 s later, immediately before the previously specified reset. The original ready plane defines gravitational zero; we retained that reference and event window. There is no external event actuator, and $D_{\text{event}} = 2mgH = 5.886$ J. Independently evaluated storage is

$$U = \tfrac{1}{2}m\sum_i \dot{y}_i^2 + mg\sum_i y_i + \tfrac{1}{2}I\dot{\phi}^2 + 25\phi^2 + \tfrac{1}{2}k\sum_i \delta_i^2. \tag{2}$$

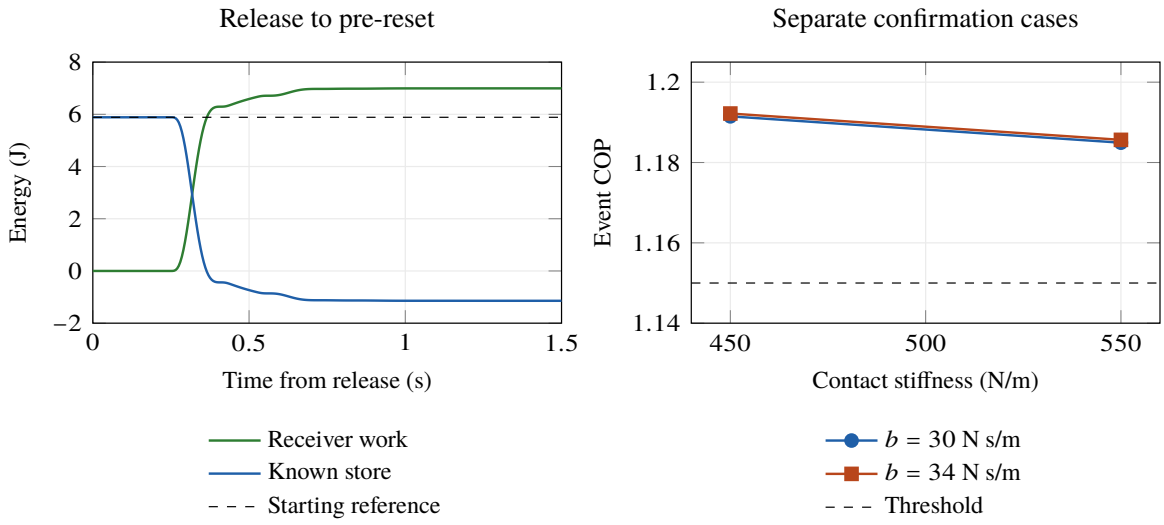



**Figure 2.** Executed lever event and separately declared confirmation cases. The store includes gravity relative to the original ready plane. Lines on the right connect the computed endpoints.

At $k = 500$ N/m and $b = 32$ N s/m, receiver work is **6.994139 J**, giving **COP 1.188267**. Bearing heat is 0.034598 J. Final storage is −1.142737 J: gravitational energy −2.244651 J plus 1.101914 J of kinetic and elastic storage. The drivers end 0.114406 m below the ready plane. Their additional descent supplies the difference from the starting reference; the independently calculated account closes numerically.

**Table 1.** Representative event and four separate confirmation cases, each executed at three solver settings.

| Contact $k$ (N/m) | Receiver $b$ (N s/m) | Useful work (J) | Event COP |
|---:|---:|---:|---:|
| 500 | 32 | 6.994139 | 1.188267 |
| 450 | 30 | 7.013120 | 1.191492 |
| 450 | 34 | 7.017282 | 1.192199 |
| 550 | 30 | 6.974505 | 1.184931 |
| 550 | 34 | 6.978687 | 1.185642 |

**Make the margin visible.** A scale and ruler determine initial energy; a load cell and encoder give $W_R = \int F_R \, ds_R$. Predicted peaks are 52.40 N and 1.637 m/s, with 0.156 m receiver excursion. Assume a calibrated 200 Hz response, 2 kHz sampling and at most 0.1 ms channel skew. Including gain, offset, timing, loading and reference allowances gives $B_{\text{in}} = 0.02943$ J and $B_{\text{out}} = 0.06608$ J. The conservative COP is **1.171**, above 1.15; all four confirmation cases also pass. The predicted threshold margin is 0.225239 J. In the retained full history, reset requires another 2.210180 J gross input; its COP is 0.995077.

## 2 A third-derivative contact law

A mass, spring and resistance describe a familiar impact approximation. A higher-order extension adds an internal force proportional to jerk, the rate of change of acceleration. Dimensional construction permits this term; its physical realization is the open question. It gives a precise way to ask what resistance work the assumed law predicts before the moving mass stops.

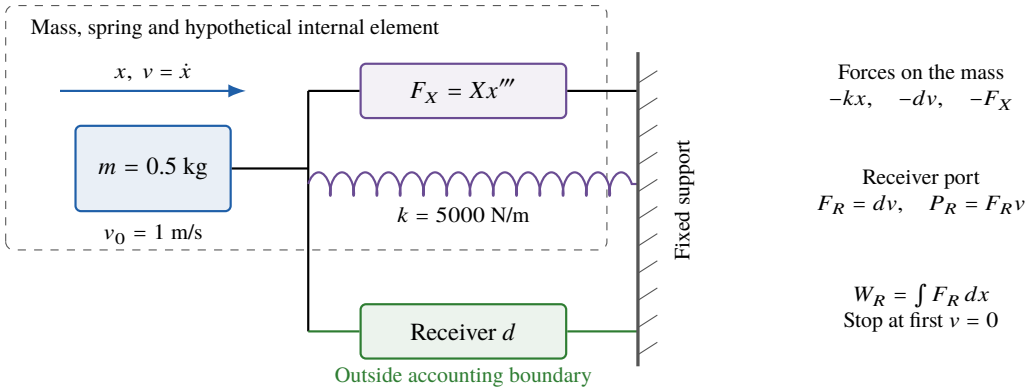



**Figure 3.** Parallel mechanical branches share displacement $x$ relative to a fixed support; the receiver lies outside the dashed accounting boundary. The element port force is $F_X$, and its force on the mass is $-F_X$. Schematic; internal storage and preparation remain open.

Let displacement be $x$, velocity $v$, mass $m$, spring stiffness $k$ and receiver resistance $d$. Add the constitutive force $F_X = Xx'''$, with $X = \alpha md/k$ in kg s. Newton's law becomes

$$Xx''' + mx'' + dx' + kx = 0, \qquad P_R = dv^2, \qquad P_X = Xx'''v. \tag{3}$$

The element exerts $-F_X$ on the mass; $P_X = F_X v$ is positive into the element. This force-work is computed independently of the kinetic and spring stores.
At $\alpha = 1$, the operator factors as $(dD+k)((m/k)D^2+1)$, where $D$ means time differentiation. For positive $m, d, k, X$, the stability condition $md > Xk$ is equivalent to $\alpha < 1$. Root stability and physical realizability are separate questions.

### 2.1 First contact to first stop

We used $m = 0.5$ kg, $k = 5000$ N/m, $x(0) = 0$, $v(0) = 1$ m/s and $x''(0) = 0$. The event ends at the first positive-time zero of $v$. No external forcing is specified. Prepared kinetic energy is 0.250 J; resistance work is the useful output. Permanent nail penetration is outside this spring-resistance model.

For $\alpha = 1$, the independent exact reference is

$$\omega = \sqrt{k/m}, \quad x = \frac{v_0}{\omega}\sin\omega t, \quad t_* = \frac{\pi}{2\omega}, \quad \frac{W_R}{mv_0^2/2} = \frac{\pi d}{2\sqrt{mk}}. \tag{4}$$

The simulation, rather than substitution alone, supplies the plotted work. At $d = 50$ N s/m, it reaches the first stop in 15.708 ms, after 10.000 mm travel. Receiver work is **0.392699 J**, and **COP is 1.570796**.

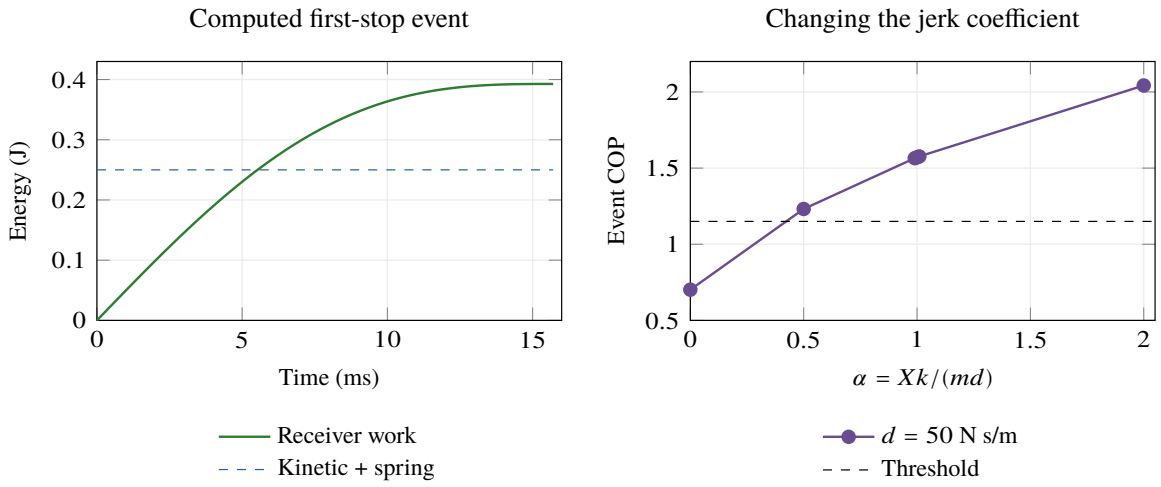



**Figure 4.** Executed first-stop work and derivative-coefficient comparison. At the reference point the kinetic-plus-spring sum remains 0.250 J while the receiver accumulates work.

**Table 2.** Four of the declared comparisons. All cases and independent integrations were retained.

| $\alpha$ | $d$ (N s/m) | Receiver work (J) | Event COP |
|---|---|---|---|
| 0, ordinary control | 50 | 0.175391 | 0.701564 |
| 0.5 | 50 | 0.307903 | 1.231611 |
| 1 | 50 | 0.392699 | 1.570796 |
| 2 | 50 | 0.510642 | 2.042569 |

At the reference stop, kinetic energy is zero and spring energy is 0.250 J. The known-store remainder $\Delta U + W_R$ is +0.392699 J. Independently integrated internal force-work is $W_X = -0.392699$ J. This locates the modeled transfer without assigning a hidden store or an external balancing supply. What material or controlled mechanism could implement $F_X$ is still open.

**A short, measurable pulse.** Peak receiver force is 50 N and incoming speed is 1 m/s. A load cell, displacement encoder or photogates, and a synchronous DAQ can resolve those scales. The final prospective specification uses a characterized 5 kHz force response, 40 kHz sampling and at most 1 $\mu$s skew. Zero pre-contact sensor state was included. An earlier 2 kHz response exceeded its allowance in some cases and was retained as a failed instrumentation comparison.

With input bound 0.005 J and output bound 0.00841 J, conservative COP is **1.507**. Four new confirmation cases at $\alpha = 0.9, 1.1$ and $d = 45, 55$ N s/m give COP 1.36188–1.79145 and pass the same criteria. Across both stages, all 14 declared configurations were executed with three solver settings. Preparation of the acceleration state and the internal component's energy account remain concrete subjects for the next investigation.

## 3 A signed-residue electrical port

A terminal impedance can hide several internal relaxation modes. Positive residues admit familiar resistor-capacitor constructions. Signed residues extend the mathematical family: some combinations remain positive real, while others predict delivery from the internal port. The useful question is whether that specified law can deliver work to an actual modeled receiver.

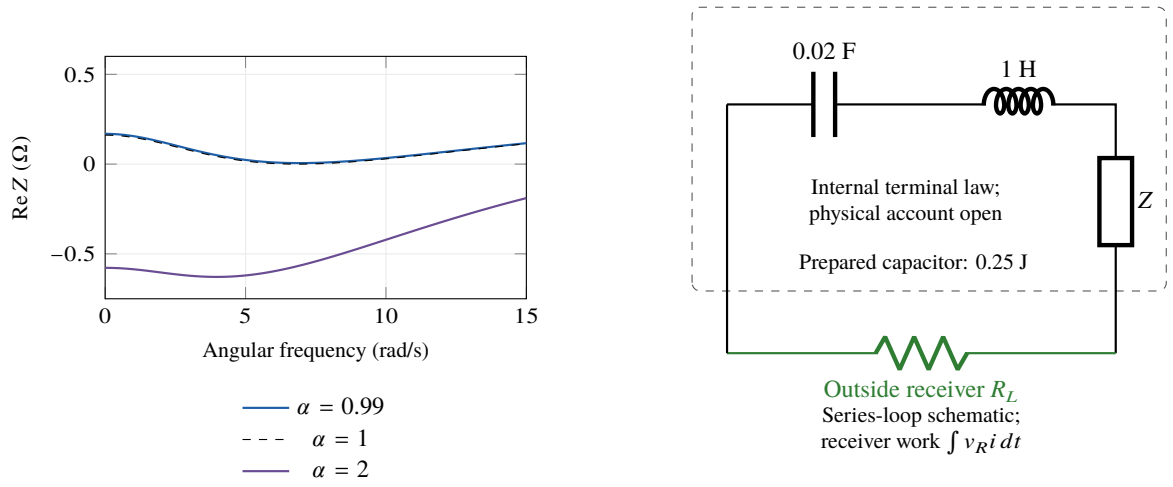



**Figure 5.** The signed-residue impedance and its declared loaded extension. The negative-real region grows with the residue scale. The boundary diagram identifies the prepared store and outside receiver.

The two-state terminal law is

$$\begin{aligned} \tau_j \dot{z}_j &= i - z_j, \qquad v_X = 0.35i + R_1 z_1 + 0.55 z_2, \\ (\tau_1, \tau_2) &= (0.08, 0.18)\ \text{s}, \qquad R_1 = -0.738869257399\alpha\ \Omega. \end{aligned} \tag{5}$$

Thus $Z(s) = 0.35 + R_1/(1 + 0.08s) + 0.55/(1 + 0.18s)$ ohm. The numerical boundary scale is $\alpha = 1$; slightly below it the real part is nonnegative at every frequency. We retained that control and extended the explicitly signed law into its active region. This is a conditional terminal model, with its internal component stores and physical implementation still unspecified.

### 3.1 Delivering work into a load

Connect this port in series with $L = 1$ H, $C = 0.02$ F and an external receiver $R_L$. Kirchhoff's laws give

$$\dot{q} = i, \qquad L\dot{i} = -q/C - R_L i - v_X, \qquad P_R = R_L i^2, \qquad P_X = v_X i. \tag{6}$$

Initially $q = 0.100$ C, $i = 0$ and $z_1 = z_2 = 0$. The prepared capacitor contains 0.250 J. The event is the declared 0–10 s interval, ending before any disconnection; no reset or external event supply is specified. Known electrical storage is $U = q^2/(2C) + Li^2/2$. The internal port's physical energy account remains open rather than being assigned zero.

We executed six residue scales $\alpha = 0.99, 1, 1.01, 1.2, 1.5, 2$ and three receivers $R_L = 0.01, 0.05, 0.1\ \Omega$. The representative $\alpha = 2$, $R_L = 0.1\ \Omega$ case gives **4.480110 J** useful work and **event COP 17.920440**.

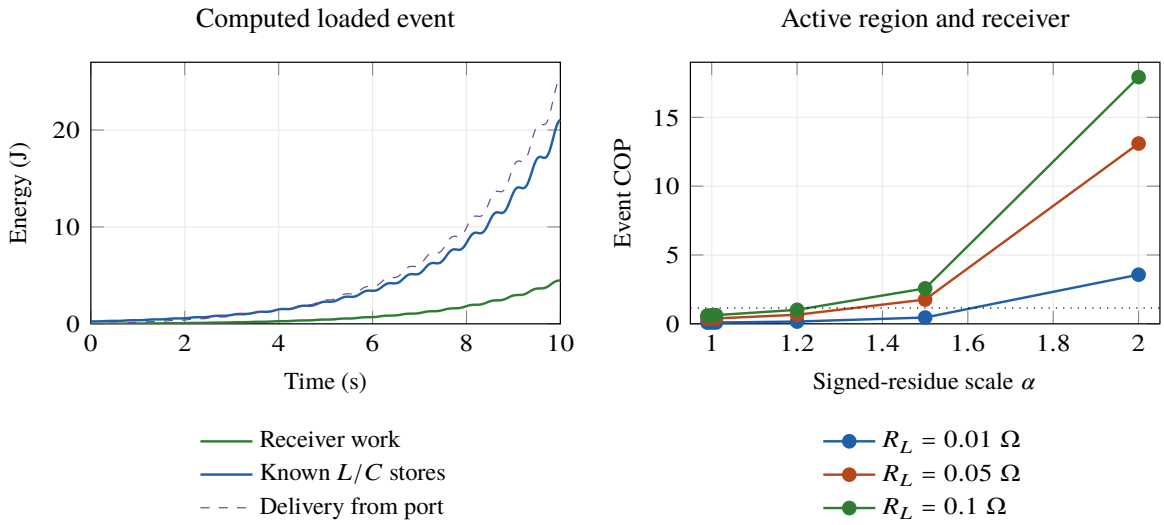



**Figure 6.** Computed work, independent known stores and internal-port delivery; receiver comparisons use the same 10 s event. The dotted line on the right is COP 1.15.

**Table 3.** Executed residue comparisons at $R_L = 0.1\ \Omega$. Remaining storage is separate from useful output.

| Residue scale $\alpha$ | Receiver work (J) | Final $L/C$ store (J) | Event COP |
|---:|---:|---:|---:|
| 0.99 | 0.150541 | 0.086090 | 0.602163 |
| 1 | 0.153998 | 0.090979 | 0.615992 |
| 1.5 | 0.643562 | 1.395023 | 2.574247 |
| 2 | 4.480110 | 21.075454 | 17.920440 |

The representative trajectory ends with 20.678461 J in the capacitor and 0.396993 J in the inductor. Direct integration gives $W_X = -25.305564$ J. Consequently $\Delta U + W_R + W_X$ is numerically small, while the partial-store remainder $\Delta U + W_R$ is 25.305564 J. Both values matter: the calculation resolves the specified terminal transfer, and leaves the internal physical account for investigation.

The connected circuit has an unstable pole pair with real part $+0.221483\,\mathrm{s}^{-1}$. Its growing trajectory makes the reported COP specific to the declared 10 s event and the law's assumed signal range.

**Signals for the bench.** The receiver reaches 0.603 V and 6.032 A; capacitor voltage reaches 45.474 V. A calibrated shunt, differential voltage channels and synchronized DAQ can integrate the actual receiver voltage-current product. Assume 200 Hz characterized channels, 2 kHz acquisition and 0.1 ms maximum skew. Input bound 0.005 J and output bound 0.09067 J leave conservative COP **17.213**. The threshold margin is 4.192610 J. Four separate cases at $\alpha = 1.8, 2.2$ and $R_L = 0.08, 0.12\ \Omega$ give COP 7.24467–43.73809 and pass the same checks. All 22 configurations were executed with three solver settings. The next physical step is to identify a port realization and measure its own energy transfers alongside those already specified.

## 4 When an impedance fit delivers work

A model can fit terminal measurements while changing the work it predicts. This investigation starts with a specified passive reference and an unconstrained four-pole approximation. The approximation develops a negative real impedance band. Connecting both laws to the same prepared circuit turns that spectral difference into a measurable work comparison.

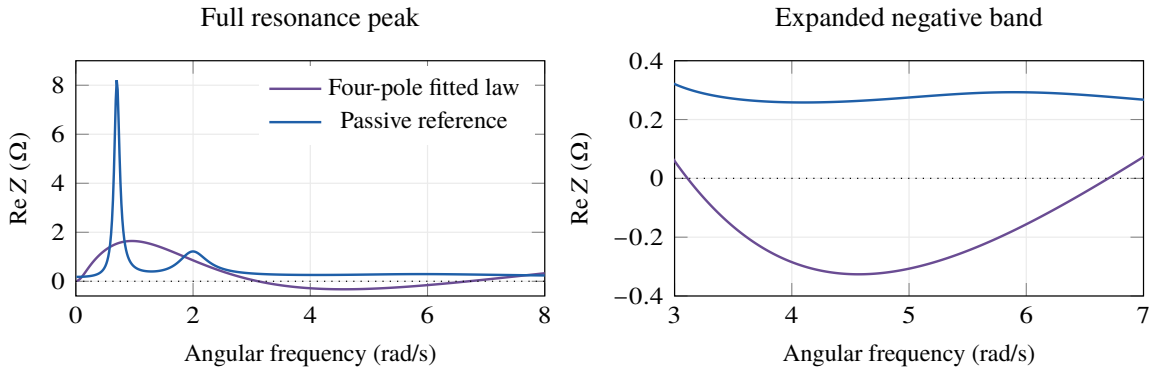



**Figure 7.** Calculated real impedance of the fitted law and its passive reference. Negative values belong to the fitted law; they are not a property transferred to the reference.

The reference has three parallel RLC branches in series with 0.18 Ω:

$$Z_{\mathrm{ref}}(s) = 0.18 + \sum_{j=1}^{3} \frac{a_j s}{s^2 + 2\zeta_j \omega_j s + \omega_j^2}, \tag{7}$$

with $a = (0.90, 0.60, 0.35)$ Ω/s, $\zeta = (0.08, 0.15, 0.30)$ and $\omega = (0.7, 2, 6)$ rad/s. Each term has nonnegative real part. An unconstrained real least-squares fit on 321 logarithmic frequencies from 0.03 to 4 rad/s uses poles $p = (-0.25, -0.8, -2.5, -8)$ s$^{-1}$ and gives

$$\dot{z}_j = p_j z_j + i, \qquad v_X = D_0 i + \sum_j c_j z_j, \qquad D_0 = 2.85295645\ \Omega, \tag{8}$$

where $c = (-0.24049188, -2.08682151, 17.41370233, -49.90285714)$ in Ω/s. Calculations retained full precision. The fit's relative complex discrete $L^2$ error is 0.823897 on its fitting band. Its minimum sampled real impedance is −0.326217 Ω near 4.571 rad/s. This gives a concrete prediction to challenge, with the supplied passive reference as an independent control.

### 4.1 One fixture, two terminal laws

We used the series circuit of (6), with $L = 1$ H, variable $C$ and $R_L$, 0.250 J initially in the capacitor, zero current and zero internal states. Both laws evolve for the same declared 10 s event, without an external event supply. The fitted states have no assigned physical stores; its terminal power $v_X i$ is integrated directly. The reference also has independently evaluated branch stores and resistor losses.

For $C = 0.05$ F and $R_L = 0.1$ Ω, the fitted law gives **0.419981 J** receiver work and **COP 1.679922**. The matched reference delivers **0.061818 J**, giving **COP 0.247271**. The difference is a result about these two model laws. It cannot establish above-one performance for the passive reference.

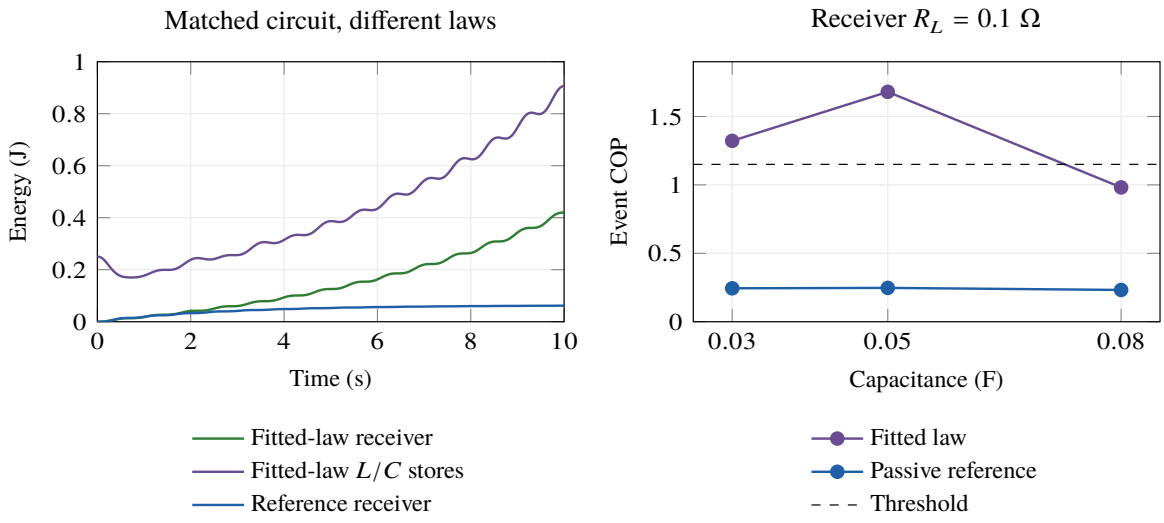



**Figure 8.** Executed work and capacitance comparisons. The same initial energy and receiver are used for both terminal laws.

**Table 4.** Actual loaded calculations at $R_L = 0.1\ \Omega$, including a below-threshold fitted case and the matched reference.

| Law | $C$ (F) | Receiver work (J) | Event COP |
|---|---|---|---|
| Fitted | 0.03 | 0.330435 | 1.321742 |
| Fitted | 0.05 | 0.419981 | 1.679922 |
| Fitted | 0.08 | 0.245376 | 0.981502 |
| Reference | 0.05 | 0.061818 | 0.247271 |

The representative fitted case leaves 0.907188 J in its known $L/C$ stores and gives $W_X = -1.077168$ J by direct terminal-power integration. The reference ends with 0.010018 J total storage and exports 0.178164 J through its other resistors. These separate endpoint accounts explain exactly which transfers each law specifies.

Although the fitted port's poles are negative, the connected circuit has an unstable pair with real part $+0.086147\,\mathrm{s}^{-1}$. The reported COP samples its growing trajectory over 10 s, assuming the law holds at these signals.

**Measure the disagreement.** Predicted fitted-case peaks are 0.1285 V at the receiver, 1.285 A, and 6.014 V across the capacitor. The same 200 Hz, 2 kHz and 0.1 ms prospective acquisition requirements used above apply. Input bound 0.005 J and output bound 0.008958 J give conservative COP **1.612**, with a 0.132481 J threshold margin. Four separately declared fitted cases at $C = 0.04, 0.06$ F and $R_L = 0.08, 0.12\ \Omega$ give COP 1.28211–1.83204 and pass. All 13 circuits were executed for both laws and three solver settings. Measure a realized port's terminal response and energy history together: that decides which law describes the apparatus.

## A common event and measurement account

Each model's event, denominator and receiver were fixed before evaluating work:

$$D_{\text{event}} = E_{\text{start}} + \sum_k \int_{t_0}^{t_1} [P_{k,\text{external}}]_+ \, dt, \qquad \text{COP}_{\text{event}} = \frac{W_{\text{useful}}}{D_{\text{event}}}. \tag{9}$$

Count prepared stores once and gross inward work at each specified external source; report returns separately. Net receiver work subtracts backflow (zero here). The input floor is $10^{-9}$ J; all representative events have zero specified external input.

**Table 5.** Four conditional model predictions. Conservative ratios use simultaneously applicable prospective input and output bounds.

| Model | Input (J) | Useful work (J) | Event COP | Conservative COP |
|---|---|---|---|---|
| Lever | 5.886 | 6.994139 | 1.188267 | 1.171 |
| Third derivative | 0.250 | 0.392699 | 1.570796 | 1.507 |
| Signed residue | 0.250 | 4.480110 | 17.920440 | 17.213 |
| Fitted port | 0.250 | 0.419981 | 1.679922 | 1.612 |

**Independent work and storage.** For a specified account, compute $r_E = U(t_1) - U(t_0) - \int \sum_k P_k \, dt$, with powers positive inward. Evaluate endpoint stores from states; the lever includes gravity and elastic stores. Integrate each internal force-velocity or voltage-current product independently. Unknown component accounts and partial-store remainders stay distinct from numerical residuals. Preparation efficiency, reset and repeatability require separate accounts.

**Numerical evidence.** The 201 integrations cover 67 configurations, including passive and below-threshold controls. DOP853/Radau, step refinement and accepted-step 8/16-point Gaussian quadrature check the trajectories and work. Matrix exponentials check the linear families; the cubic also matches its exact first-stop solution. Acceptance required COP spread below $10^{-5}$; quadrature and accounted-energy residuals below $10^{-6}$ J for the lever, $10^{-7}$ J for the cubic, and $10^{-7}$ relative to the largest accounted energy for the electrical cases.

A separate implementation of all 67 configurations found maximum absolute disagreements of $3.63 \times 10^{-12}$ in COP, $1.82 \times 10^{-12}$ J in work quadrature, and $6.26 \times 10^{-13}$ J in endpoint stores. Its largest account residual was $3.14 \times 10^{-12}$ J: the full lever store account, or known mechanical/electrical stores with independently integrated internal-port work. These numerical estimates are not certified bounds; physical component-law uncertainty remains open. Earlier sensor-response failures were retained and followed up.

**What the instrument bounds mean.** Lever input uncertainty is 0.5%; the other initial-energy bounds are 2%. Lever output allowances cover 0.3% force gain, 0.2% displacement gain, 0.02 N force offset, sampled timing errors, 0.2% bandwidth/loading and 0.01 J resolution/boundary allowance. Other outputs allocate 0.5% each to the two measured gains, bandwidth/loading and timing/ sampling, plus 0.5 mJ offset/resolution. Gain errors are composed multiplicatively. Filtering and synchronization controls fit these allocations. For simultaneous bounds $B_{\text{in}}, B_{\text{out}}$ we require $(W_{\text{useful}} - B_{\text{out}})/(D_{\text{event}} + B_{\text{in}}) > 1.15$. The assumed calibrations must be demonstrated on an apparatus. They establish a concrete measurement target for each conditional prediction.